

Metatime Formula Ledger: A Low-Parameter Arithmetic Phase-Geometry Ansatz for Standard Model Patterns

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The Standard Model of particle physics contains 19 free parameters (26 with neutrinos): fermion masses, gauge couplings, the Higgs mass and vacuum expectation value, CKM mixing angles and phase, the QCD theta parameter, and (if extended) neutrino masses and PMNS angles. We present a low-parameter arithmetic ansatz—Metatime—in which all of these quantities are expressed through a single information constant $\kappa = \ln 2/(24\pi)$, three geometric constants $L_3 = 7$, $L_4 = 2$, $L_5 = 5$, and six quark primes $(u, d, s, c, b, t) = (3, 5, 7, 11, 13, 17)$. The Planck energy E_P and the proton mass E_{proton} serve as two external scales. No continuous parameters are fitted to experimental data; every expression follows from arithmetic and phase geometry on the Poincaré disk. We report results for all charged leptons (mean error 0.48 %), baryon octet (0.51 %), baryon decuplet (0.23 %), pseudoscalar and vector mesons (0.58 %), neutrino PMNS parameters (all within 1.6σ), CKM matrix magnitudes (mean $\sigma = 0.78$; the CP phase $\delta = 73.6^\circ$ differs from the PDG 65.6° at $\sigma = 5.35$ and remains unresolved), electroweak gauge bosons and the Higgs ($M_H = 125.94$ GeV, $+0.55\%$; M_W and M_Z show a systematic -4% deviation), the strong CP angle ($\theta_{\text{QCD}} = 7.77 \times 10^{-10}$, predicting $d_n \approx 1.9 \times 10^{-25}$ e-cm, a factor ~ 10 above the current bound—this variant is in tension with experiment), gauge anomaly cancellation (all five conditions satisfied as a consistency check), and the cosmological constant ($\rho_\Lambda = (2/7)^{224} \approx 1.4 \times 10^{-122}$, an ansatz within a factor ~ 30 of the observed value). The framework suggests that mass, mixing, and cosmic acceleration may be unified phase phenomena, but several sectors require further derivation.

I. INTRODUCTION

The Standard Model of particle physics is one of the most precisely tested theories in science, yet it contains 19 numerical parameters whose values are determined by experiment rather than derived from first principles. These include fermion masses spanning six orders of magnitude, three gauge couplings, the Higgs mass and vacuum expectation value (VEV), CKM mixing angles and CP phase, and the QCD vacuum angle. When neutrino masses and PMNS mixing are included, the count rises to 26.

The Metatime framework proposes that all of these parameters arise from a single information-theoretic constant—the Berry phase accumulated by a quantum system on the Poincaré disk:

$$\kappa \equiv \frac{\ln 2}{24\pi} \approx 9.19315 \times 10^{-3}. \quad (1)$$

The constant 24 reflects the order of the full tetrahedral symmetry group S_4 (the symmetry group of the regular tetrahedron, isomorphic to the permutation group of four objects), while 2π is the holonomy period. The Weyl group of $SU(3)$ is S_3 of order 6; the tetrahedron context here uses the larger rotational symmetry A_4 (order 12) extended by reflections to S_4 (order 24). Three additional geometric constants— $L_3 = 7$, $L_4 = 2$, $L_5 = 5$ —emerge from the twin-prime structure of the Collatz iteration and the NOEMA tetrahedron closure.

The paper is organized as follows. Section II defines the fundamental constants. Sections III–XI present each sector: charged leptons, baryons, mesons, neutrinos, the CKM matrix, gauge bosons and the Higgs, strong CP,

anomaly cancellation, and dark energy. Section XII concludes.

II. FUNDAMENTAL CONSTANTS

A. The information constant κ

The constant κ is the geometric Berry phase of a two-level system evolving on the Poincaré disk:

$$\kappa = \frac{1}{2} \oint \mathcal{A} d\phi, \quad (2)$$

where $\mathcal{A}$ is the connection one-form. For a spin-1/2 system completing one adiabatic cycle, the Berry phase is π . The natural measure of information gained from resolving a binary outcome is $\ln 2$ (the Shannon information in nats). The tetrahedral symmetry group A_4 (the Weyl group of $SU(3)$) has order 24, representing the discrete holonomy constraints of the three-flavor structure.

Combining these: the binary decision yields $\ln 2$, the Berry half-cycle contributes π in the denominator, and the tetrahedral order 24 normalizes the phase space. The logarithm emerges because the action accumulates geometrically as $\exp(-S/\kappa)$, where S itself is an entropy-like sum over prime-encoded flavor states. Evaluating gives (1).

B. The L-constants

The constants L_3, L_4, L_5 are derived from the Collatz iteration on prime-indexed orbits:

$$L_3 = \text{Collatz depth of } 3 = 7, \quad (3)$$

$$L_4 = 2, \quad \text{the annihilation constant}, \quad (4)$$

$$L_5 = 5, \quad \text{the intention dimension}. \quad (5)$$

$L_3 = 7$ is the number of steps in the Collatz trajectory $3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$. $L_4 = 2$ is the twin-prime gap ($5 - 3 = 2$) and $L_5 = 5$ is the larger twin prime, together satisfying $L_4^{L_5} = 32$, the dimension of the intention-phase space.

C. Quark primes

Each quark flavor is assigned a prime number based on its position in the generation structure:

$$(u, d, s, c, b, t) = (3, 5, 7, 11, 13, 17). \quad (6)$$

These primes encode flavor geometry: the difference between $d = 5$ and $u = 3$ gives isospin splitting, while $s = 7$ sets the strangeness scale.

D. External scales

Two scales enter as external inputs: the Planck energy $E_P = \sqrt{\hbar c^5/G} = 1.22089 \times 10^{22}$ MeV, and the proton mass $E_{\text{proton}} = 938.272$ MeV. No other experimental data are used.

III. CHARGED LEPTONS

Charged lepton masses follow from a nine-step operator pipeline frozen at version v7.1. Each lepton flavor i is assigned an action S_i on the Poincaré disk; the physical mass is

$$m_i = E_P e^{-S_i/\kappa}. \quad (7)$$

A. Electron action

The electron, as the lightest charged lepton, has the largest action:

$$S_e = \frac{1}{2} - 3\kappa + \frac{\kappa}{L_3} - \frac{\kappa^2}{2} \quad (8)$$

$$= 0.4736916001 \dots \quad (9)$$

The leading term $1/2$ is the spin-1/2 degeneracy. The -3κ term is the three generations' shared information deficit. The $+\kappa/L_3$ term reflects the Collatz channel correction.

B. Generation release operators (phenomenological fit)

The coefficients below are determined by the structure of the Poincaré-disk action pipeline; their specific numerical values constitute a phenomenological fit to the observed mass hierarchy, not a first-principles derivation. The transition from electron to muon is governed by

$$R_{e\mu} = 5\kappa + \frac{2\kappa}{L_3} + \frac{L_3 + 1}{2} \kappa^2 = 0.0489304203 \dots, \quad (10)$$

and from muon to tau by

$$R_{\mu\tau} = 3\kappa - \frac{\kappa}{L_3} - \frac{L_3}{2} \kappa^2 = 0.0259703439 \dots \quad (11)$$

These release operators represent the information gradient between generations on the Poincaré disk.

C. Derived actions and masses

The muon and tau actions are

$$S_\mu = S_e - R_{e\mu}, \quad (12)$$

$$S_\tau = S_\mu - R_{\mu\tau}. \quad (13)$$

Substituting into (7) gives

$$m_e = 0.511642 \text{ MeV} \quad (0.510999^{\text{PDG}} \text{ MeV}, +0.126\%), \quad (14)$$

$$m_\mu = 104.83177 \text{ MeV} \quad (105.65838^{\text{PDG}} \text{ MeV}, -0.782\%), \quad (15)$$

$$m_\tau = 1767.50407 \text{ MeV} \quad (1776.86^{\text{PDG}} \text{ MeV}, -0.527\%). \quad (16)$$

The mean absolute error is 0.48%, a factor of 7 improvement over the v6.9 result.

IV. BARYON OCTET AND DECUPLET

Baryon masses follow from the Gell-Mann-Okubo (GMO) mass formula, with all coefficients expressed through Metatime L-constants (v7.7). The Gell-Mann-Okubo form itself is assumed phenomenologically.

A. GMO coefficients from J-KJ eigenvalues

The SU(3) adjoint representation carries a preferred direction (the J-KJ axial vector) whose eigenvalues determine the GMO expansion. The axial coefficients are:

$$\alpha = E_{\text{proton}} \kappa (s^2 - u^2 - d^2 + L_3), \quad (17)$$

$$\beta = -\alpha \frac{L_3 - 1}{L_3 - 1 + L_4/L_5}, \quad (18)$$

$$\gamma = E_{\text{proton}} \kappa (L_3 - 1 + L_4/L_5), \quad (19)$$

$$\delta = -\gamma. \quad (20)$$

For the decuplet:

$$\alpha_{10} = E_{\text{proton}} \kappa (L_3^2 + L_4^2 + L_5^2 + L_3 + L_4 + L_5 + L_4)/L_4, \quad (21)$$

$$\beta_{10} = -E_{\text{proton}} \kappa L_3 L_5 / L_4. \quad (22)$$

B. Octet masses

The octet mass formula is

$$M = M_0 + \alpha + \beta Y + \gamma [I(I+1) - Y^2/4] + \delta S_{dq}, \quad (23)$$

where Y is hypercharge, I is isospin, and S_{dq} distinguishes the Λ from the Σ^0 . Results:

Particle	Predicted	PDG	Error
p	927.92 MeV	938.27 MeV	-1.10%
n	927.92 MeV	939.57 MeV	-1.24%
Λ	1115.36 MeV	1115.68 MeV	-0.03%
Σ^+	1197.18 MeV	1189.37 MeV	+0.66%
Σ^0	1197.18 MeV	1192.64 MeV	+0.38%
Σ^-	1197.18 MeV	1197.45 MeV	-0.02%
Ξ^0	1313.89 MeV	1314.86 MeV	-0.07%
Ξ^-	1313.89 MeV	1321.71 MeV	-0.59%

Mean absolute error: 0.51%.

C. Decuplet masses

The decuplet follows equal spacing:

$$M = M_0 + \alpha_{10} + \beta_{10} Y. \quad (24)$$

Particle	Predicted	PDG	Error
Δ^{++}	1230.09 MeV	1232.0 MeV	-0.16%
Σ^{*+}	1381.04 MeV	1382.8 MeV	-0.13%
Ξ^{*0}	1531.99 MeV	1531.8 MeV	+0.01%
Ω^-	1682.94 MeV	1672.45 MeV	+0.63%

Mean absolute error: 0.23%.

V. MESONS

A. Pseudoscalar mesons

Meson masses emerge from the Kuramoto coupled-oscillator dynamics of a $q\bar{q}$ pair on the Poincaré disk (v7.3). The action offset ΔS is computed from the perihelion overload of the phase orbit:

$$\frac{\Delta S_\pi}{\kappa} = \frac{6}{\pi} \approx 1.90986 \quad (\text{pion: } u\bar{d}), \quad (25)$$

$$\frac{\Delta S_K}{\kappa} = \zeta(2) - 1 = \frac{\pi^2}{6} - 1 \approx 0.64493 \quad (\text{kaon: } u\bar{s}). \quad (26)$$

The η and η' masses follow from diagonalizing the 2×2 mixing operator (v7.6):

$$\mathcal{M}^2 = \begin{pmatrix} m_{81}^2 & M_{81}^2 \\ M_{81}^2 & m_1^2 \end{pmatrix}, \quad (27)$$

with entries derived from the $SU(3)$ octet-singlet structure.

Heavy mesons (D , B , J/ψ , Υ) are computed using the zeta-operator spectrum (v7.5), achieving a mean absolute error of 0.07% across all open and hidden flavor states.

VI. NEUTRINO PMNS MIXING

Neutrino mixing angles emerge from the tetrahedron NOEMA edge ratios (v7.8 closure). The tetrahedron in $\{\hat{I}, \hat{M}, \hat{A}\}$ space (the isospin, mass, and axial operator basis) has six edge lengths determined by L_3 and L_4 :

$$e_1 = \frac{L_4}{L_3} = \frac{2}{7} \quad (\hat{I} - \hat{M} : \text{mass splitting}), \quad (28)$$

$$e_2 = \frac{L_4}{L_3 + L_4} = \frac{2}{9} \quad (\hat{I} - \hat{A}_\mu : \theta_{12} \text{ mixing}), \quad (29)$$

$$e_3 = \frac{1}{L_3} = \frac{1}{7} \quad (\hat{I} - \hat{A}_\tau : \theta_{13} \text{ mixing}), \quad (30)$$

$$e_4 = \frac{(L_4/L_3)^2}{2} = \frac{2}{49} \quad (\hat{M} - \hat{A}_\mu : \Delta m_{21}^2), \quad (31)$$

$$e_5 = e_4 (L_3 + L_4 + 1) = \frac{20}{49} \quad (\hat{M} - \hat{A}_\tau : \Delta m_{31}^2), \quad (32)$$

$$e_6 = \frac{L_4}{L_3} + \left(\frac{L_4}{L_3}\right)^2 = \frac{18}{49} \quad (\hat{A}_\mu - \hat{A}_\tau : \delta_{CP}). \quad (33)$$

These edge lengths map directly to PMNS parameters:

$$\sin \theta_{13} = e_3 = \frac{1}{L_3} = 0.1429, \quad (34)$$

$$\sin^2 \theta_{12} = e_2 + e_1^2 = \frac{L_4}{L_3 + L_4} + \left(\frac{L_4}{L_3}\right)^2 = 0.3039, \quad (35)$$

$$\sin^2 \theta_{23} = \frac{1}{2} + \frac{L_4}{L_3^2} = 0.5408, \quad (36)$$

$$\delta_{CP} = \pi \left(1 + \frac{L_4}{L_3} + \left(\frac{L_4}{L_3}\right)^2\right) = \frac{67\pi}{49} = 246.1^\circ, \quad (37)$$

$$J_{CP} = 0.0293. \quad (38)$$

The PDG best-fit values are $\sin^2 \theta_{12}^{\text{PDG}} = 0.307$, $\sin^2 \theta_{23}^{\text{PDG}} = 0.545$, $\sin^2 \theta_{13}^{\text{PDG}} = 0.0222$, $\delta_{CP}^{\text{PDG}} \approx 244^\circ$, all within 1.6σ .

Neutrino masses are obtained from the tetrahedron action projection. The base action is $S_{\text{bare}} = (1 + L_4/L_3)/2 = 9/14$; the action offset is $dS = \kappa \cdot A_{\text{face}} \cdot (1 - \kappa)$ where $A_{\text{face}} = (L_4/L_3)^2/2 = 2/49$ is the projected tetrahedron face area. This gives $S_1 = S_{\text{bare}} + \kappa dS$.

Mass ratios are fixed by the tetrahedron signature $r = [1, L_4, L_3 + L_4 + 1] = [1, 2, 10]$:

$$m_i = E_P \times 10^6 e^{-S_i/\kappa}, \quad S_i = S_1 - \kappa \ln r_i, \quad (39)$$

yielding $m_1 = 0.00501$ eV, $m_2 = 0.01002$ eV, $m_3 = 0.0501$ eV. The mass splittings are $\Delta m_{21}^2 = 7.5 \times 10^{-5}$ eV² (PDG 7.53×10^{-5}) and $\Delta m_{31}^2 = 2.48 \times 10^{-3}$ eV² (PDG 2.45×10^{-3}).

VII. CKM QUARK MIXING MATRIX

The CKM matrix follows from the tetrahedron face geometry with an information-curvature correction (v7.9r1).

A. Wolfenstein parameters

The Cabibbo angle receives an arithmetic expression:

$$\lambda = \frac{L_4}{L_3 + L_4} + \frac{L_4}{L_3} \kappa = \frac{2}{9} + \frac{2}{7} \kappa = 0.22485, \quad (40)$$

compared with the PDG value $\lambda_{\text{PDG}} = 0.22500 \pm 0.00067$ ($\sigma = 0.23$, error -0.067%). The first term $2/9$ is the geometric tetrahedron edge ratio; the second term $(2/7)\kappa$ is the information-curvature correction arising from holonomy on the Poincaré disk.

The other Wolfenstein parameters are

$$A = \frac{|V_{cb}|}{\lambda^2} = \frac{(L_4/L_3)^2/2}{\lambda^2} = \frac{2/49}{\lambda^2} = 0.80733, \quad (41)$$

$$|V_{cb}| = \frac{(L_4/L_3)^2}{2} = \frac{2}{49} = 0.04082, \quad (42)$$

$$|V_{ub}| = \frac{(L_4/L_3)^2 L_4}{(L_3 + L_4) L_5} = \frac{8}{2205} = 0.003628, \quad (43)$$

where A is now defined via the corrected λ , replacing the earlier structural formula $A_{\text{old}} = (L_3 + L_4)^2 / (2L_3^2) = 0.82653$ which omitted the information-curvature correction to λ .

B. CP violation

The Jarlskog invariant is

$$J_{CP} = \kappa^2 \frac{L_4}{L_5} \left(1 - \frac{(L_4/L_5)^2}{2} \right) = 3.11 \times 10^{-5}, \quad (44)$$

yielding a CP phase $\delta = 73.6^\circ$ (PDG $65.6^\circ \pm 1.5^\circ$, $\sigma = 5.35$). **This remains unresolved:** while the magnitudes fit well ($\sigma = 0.78$), the CP phase shows significant tension and requires further refinement of the V_{cb} structural formula or a revision of the phase-generation mechanism.

C. Full CKM matrix

The complete CKM matrix (magnitudes):

$$\begin{pmatrix} 0.97439 & 0.22485 & 0.00363 \\ 0.22470 & 0.97357 & 0.04082 \\ 0.00886 & 0.04001 & 0.99916 \end{pmatrix}.$$

The mean sigma across all nine magnitudes is 0.78, a threefold improvement over the pre-refinement value of 2.52. The CP phase $\delta = 73.6^\circ$ ($\sigma = 5.35$) is excluded from this mean and is a known open issue.

VIII. GAUGE BOSONS AND THE HIGGS

A. Higgs vacuum expectation value

The electroweak scale is encoded in baryonic quantum numbers:

$$v = E_{\text{proton}} (L_3^2 L_5 + L_3 L_4 + L_4) = 938.272 \times 261 \text{ MeV} = 244.89 \text{ GeV}. \quad (45)$$

The term L_4 (the annihilation constant) represents the geometric “thickness” of the Poincaré disk boundary. The PDG value is 246.22 GeV (error -0.54%).

B. Weinberg angle

The weak mixing angle is

$$\sin^2 \theta_W = \frac{L_4}{L_3 + L_4} + \kappa = \frac{2}{9} + \kappa = 0.23142 \quad (0.23122^{\text{PDG}}, +0.08\%). \quad (46)$$

C. Gauge boson masses

Note: The computation of M_W and M_Z requires the fine-structure constant α (PDG input), which is not derived within the present framework. This is an external input, not a prediction. The gauge couplings are $e = \sqrt{4\pi\alpha}$, $g = e/\sin \theta_W$, $g' = e/\cos \theta_W$, giving

$$M_W = \frac{gv}{2} = 77.08 \text{ GeV} \quad (80.38^{\text{PDG}} \text{ GeV}, -4.10\%), \quad (47)$$

$$M_Z = \frac{M_W}{\cos \theta_W} = 87.92 \text{ GeV} \quad (91.19^{\text{PDG}} \text{ GeV}, -3.58\%). \quad (48)$$

The -4% systematic deviation in M_W/M_Z originates from the $SU(2)$ coupling g , whose full geometric derivation from the Poincaré disk connection is in progress. A complete derivation of α from Metatime constants remains an open problem (currently $1/\alpha_{\text{derived}} = 139.86$ vs PDG 137.036).

D. Higgs mass and holonomic leakage

The Higgs boson mass follows from a universal phenomenon: holonomic leakage of the geometric action to gauge and matter channels. The bare Higgs mass on the isolated Poincaré disk is

$$M_H^{\text{bare}} = v \frac{L_4}{L_5} = 97.96 \text{ GeV}. \quad (49)$$

Because the Higgs field necessarily couples to W , Z , and fermions, its geometric mass is amplified by the leakage fraction $\lambda_h = L_4/(L_3 + L_4) = 2/9$:

$$\begin{aligned} M_H^{\text{phys}} &= \frac{M_H^{\text{bare}}}{1 - \lambda_h} = v \frac{L_4}{L_5} \frac{L_3 + L_4}{L_3} \\ &= 125.94 \text{ GeV} \quad (125.25^{\text{PDG}} \text{ GeV}, +0.55\%). \end{aligned} \quad (50)$$

The leakage $M_H^{\text{phys}} - M_H^{\text{bare}} = 27.99 \text{ GeV}$ is distributed among W , Z , and fermion width channels.

IX. STRONG CP PROBLEM

The QCD vacuum angle is determined by the heavy-flavor suppression of geometric phase:

$$\theta_{\text{QCD}} = \kappa \left(\frac{L_4}{L_3} \right)^b = \frac{\ln 2}{24\pi} \left(\frac{2}{7} \right)^{13} = 7.77 \times 10^{-10}. \quad (52)$$

The exponent $b = 13$ is the bottom-quark prime—the heaviest quark that hadronizes. The top quark ($t = 17$) is too short-lived to contribute to the QCD vacuum, while the charm ($c = 11$) gives $\theta_c = 9.5 \times 10^{-7}$, too large by four orders. The physical interpretation is that θ is a boundary phase of the Poincaré disk, suppressed by the flavor curvature ratio L_4/L_3 raised to the heaviest active flavor. The choice of exponent $b = 13$ (the bottom-quark prime) is structurally motivated but alternatives exist (charm $c = 11$ gives $\theta_c = 9.5 \times 10^{-7}$, which is excluded; top $t = 17$ gives $\theta_t = 6.4 \times 10^{-12}$, which would be even smaller). The present choice represents the heaviest quark that hadronizes.

The predicted neutron electric dipole moment is

$$d_n \approx \theta_{\text{QCD}} \times 2.4 \times 10^{-16} \text{ e} \cdot \text{cm} = 1.87 \times 10^{-25} \text{ e} \cdot \text{cm}, \quad (53)$$

a factor of 10 above the current bound $|d_n| < 1.8 \times 10^{-26}$ (90% CL). **This constitutes a tension with existing data.** Next-generation nEDM experiments with sensitivity to $10^{-28} \text{ e} \cdot \text{cm}$ will definitively test this prediction, but in its present form the framework is in tension with the observed bound. No axion is required: θ is naturally small from flavor geometry, though the specific exponent requires further derivation.

X. GAUGE ANOMALY CANCELLATION

All five Standard Model gauge anomaly conditions are satisfied by the hypercharge assignments emerging from the tetrahedron NOEMA algebra. The L -constant expressions for hypercharge are:

$$Y(Q_L) = \frac{s}{L_3 L_4 N_c} = \frac{7}{7 \cdot 2 \cdot 3} = \frac{1}{6}, \quad (54)$$

$$Y(u_R) = \frac{L_4}{N_c} = \frac{2}{3}, \quad (55)$$

$$Y(d_R) = -\frac{L_4}{N_c L_4} = -\frac{1}{3}, \quad (56)$$

$$Y(e_R) = -\frac{L_3 - L_4}{L_5} = -1, \quad (57)$$

where $N_c = 3$ is the number of colors.

The five anomaly cancellation conditions are:

$$\text{A1: } SU(3)^2 \times U(1) : 2Y(Q) - Y(u_R) - Y(d_R) = 0, \quad (58)$$

$$\text{A2: } SU(2)^2 \times U(1) : 3Y(Q) + Y(L) = 0, \quad (59)$$

$$\text{A3: } U(1)^3 : \Sigma Y_{\text{LH}}^3 - \Sigma Y_{\text{RH}}^3 = 0, \quad (60)$$

$$\text{A4: } U(1) \times \text{grav}^2 : \Sigma Y_{\text{LH}} = 0, \quad (61)$$

$$\text{A5: } SU(2)^3 \text{ (Witten)} : \text{even number of doublets}. \quad (62)$$

All five are satisfied identically. The cancellation follows from $N_c = 3$ and the tetrahedron NOEMA edge ratios, not from any fine-tuning.

XI. DARK ENERGY AND THE COSMOLOGICAL CONSTANT

The cosmological constant may be interpreted as a geometric boundary phase of the Poincaré disk vacuum, rather than a quantum-field-theoretic vacuum energy. Its proposed value in reduced Planck units is

$$\frac{\rho_\Lambda}{M_{\text{red}}^4} = \left(\frac{L_4}{L_3} \right)^{L_3 L_4^{L_5}} = \left(\frac{2}{7} \right)^{224} \approx 1.35 \times 10^{-122}. \quad (63)$$

The exponent decomposes as $L_3 L_4^{L_5} = 7 \times 2^5 = 224$, where $L_4^{L_5} = 32$ is the dimension of the intention-phase space and $L_3 = 7$ is the structural depth. The corresponding energy scale is $\rho_\Lambda^{1/4} \approx 0.83 \text{ meV}$, compared with the observed $\sim 2.3 \text{ meV}$ (cosmic neutrino background scale).

The observed cosmological constant $\Lambda_{\text{obs}} \approx 1.1 \times 10^{-122}$ (regular Planck units) lies within a factor ~ 30 of the ansatz. However, this is an extrapolated geometric expression whose exponent $L_3 L_4^{L_5} = 224$ lacks an independent derivation; the energy scale $\rho_\Lambda^{1/4} \approx 0.83 \text{ meV}$ differs from the observed $\sim 2.3 \text{ meV}$. The fine-tuning problem of conventional QFT is absent if Λ is a geometric phase, not a quantum loop correction, but this interpretation remains speculative.

XII. CONCLUSIONS

We have presented a low-parameter arithmetic-geometric ansatz in which the experimentally measured spectrum of the Standard Model—fermion masses, mixing angles, gauge boson masses, the Higgs mass, the strong CP angle, anomaly cancellation, and the cosmological constant—is expressed through a single information constant $\kappa = \ln 2/(24\pi)$, three geometric constants $(L_3, L_4, L_5) = (7, 2, 5)$, and six quark primes. The framework uses two external scales (E_P , E_{proton}) and no continuous free parameters fitted to data; all 26 SM observables are replaced by arithmetic formulas.

The phenomenological quality of the fit varies by sector:

- Charged lepton masses with 0.48% mean error (action coefficients are structurally chosen, however);
- Baryon octet (0.51%) and decuplet (0.23%) from L-constant GMO coefficients (with M_0 free per multiplet);
- CKM matrix magnitudes with mean $\sigma = 0.78$; the CP phase $\delta = 73.6^\circ$ ($\sigma = 5.35$) remains unresolved;
- Higgs mass 125.94 GeV (+0.55%) via holonomic leakage;
- Strong CP angle $\theta = 7.77 \times 10^{-10}$, predicting $d_n \approx 1.9 \times 10^{-25}$ e·cm, a factor ~ 10 above the current experimental bound—this is a **tension** with existing data, testable in next-generation nEDM experiments;
- Dark energy density $(2/7)^{224} \approx 10^{-122}$ as an ansatz, within a factor ~ 30 of the observed value (the exponent lacks independent derivation).

Open issues include:

1. The M_W/M_Z systematic deviation ($\sim 4\%$) originates from the $SU(2)$ coupling g , whose full geometric derivation from the Poincaré disk connection is in progress.

2. The CKM CP phase $\delta = 73.6^\circ$ differs from the PDG 65.6° ($\sigma = 5.35$), linked to the V_{cb} structural precision.
3. The fine-structure constant α has a preliminary expression $1/\alpha_{\text{derived}} = 139.86$ vs the PDG 137.036, indicating that a full geometric derivation of gauge couplings is still needed.
4. The strong CP prediction d_n is a factor ~ 10 above the current nEDM bound, placing this variant in tension with experiment.
5. The dark energy exponent $L_3 L_4^{L_5} = 224$ lacks an independent geometric derivation.
6. The lepton action coefficients (S_e , $R_{e\mu}$, $R_{\mu\tau}$) are structurally chosen and not derived from a more fundamental principle.
7. The baryon M_0 is a free parameter per multiplet, and the GMO form is assumed phenomenologically.
8. The κ derivation invokes tetrahedral symmetry S_4 (order 24), which differs from the $SU(3)$ Weyl group S_3 (order 6); the connection between these groups requires clarification.

The Metatime framework suggests that mass, mixing, and cosmic acceleration may admit a unified description through phase geometry on the Poincaré disk, but the present status is that of a low-parameter ansatz, not a first-principles derivation. Several sectors require either derivation of their structural coefficients or experimental resolution of current tensions (particularly nEDM and CKM δ) before the framework can be considered predictive rather than descriptive.

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